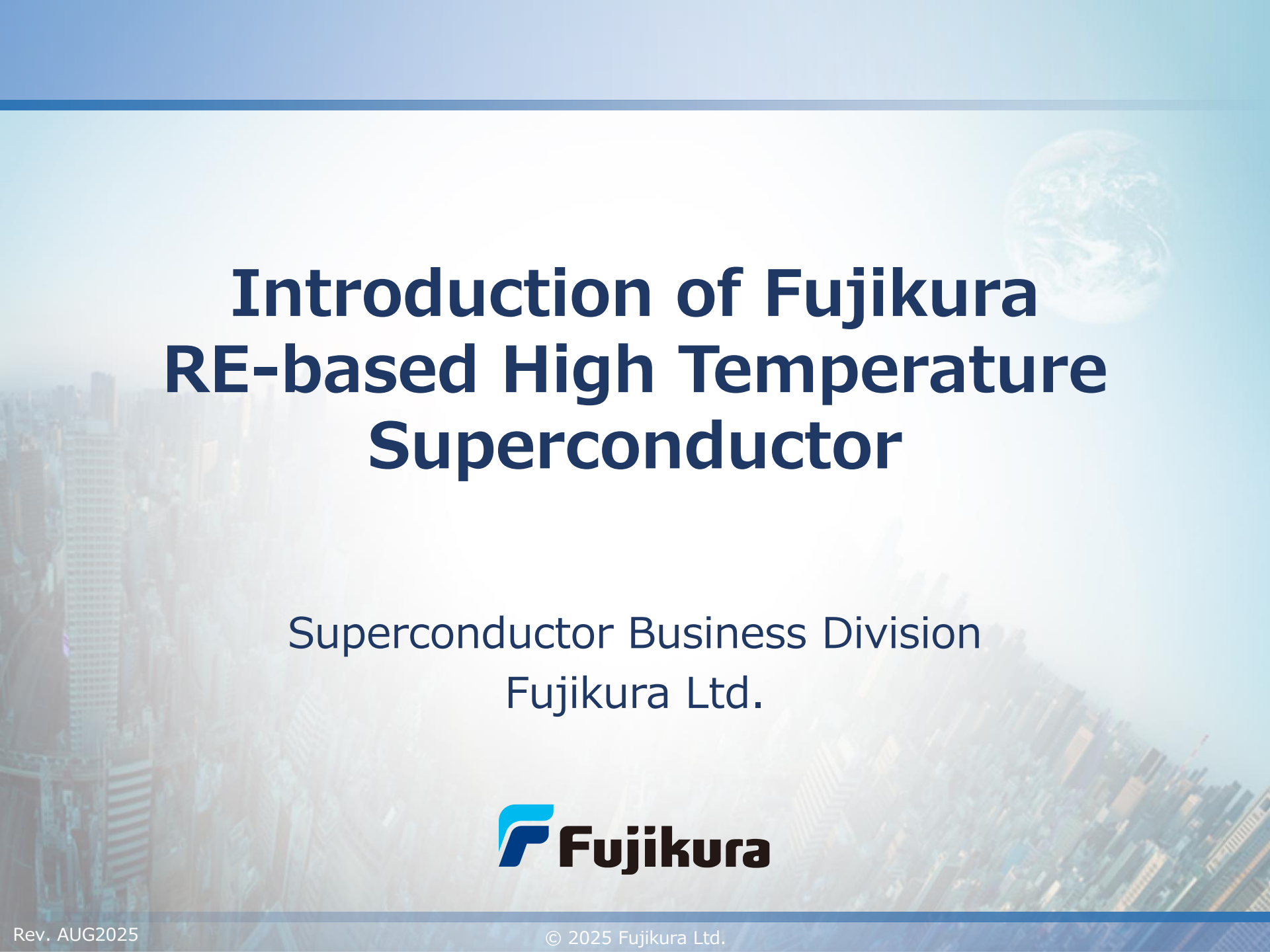




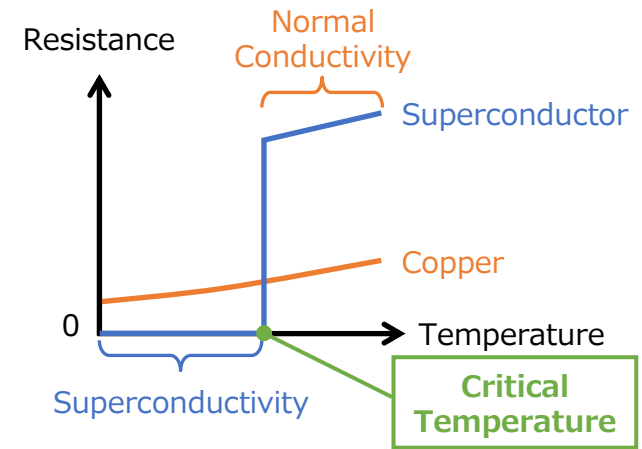
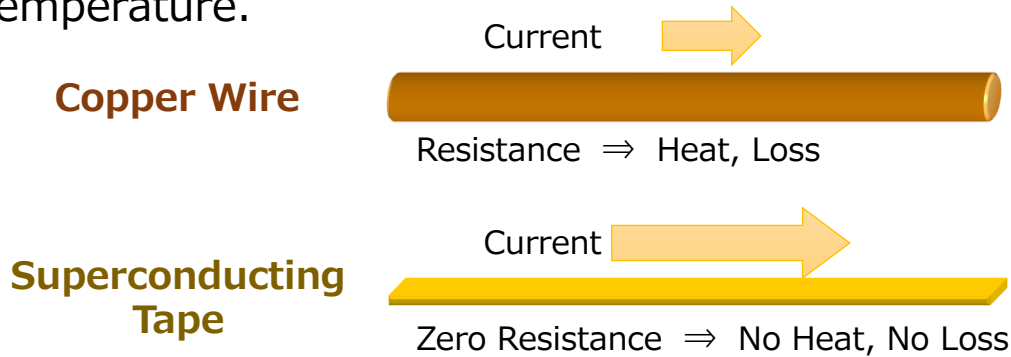
Introduction of Fujikura RE-based High Temperature Superconductor

Superconductor Business Division
Fujikura Ltd.



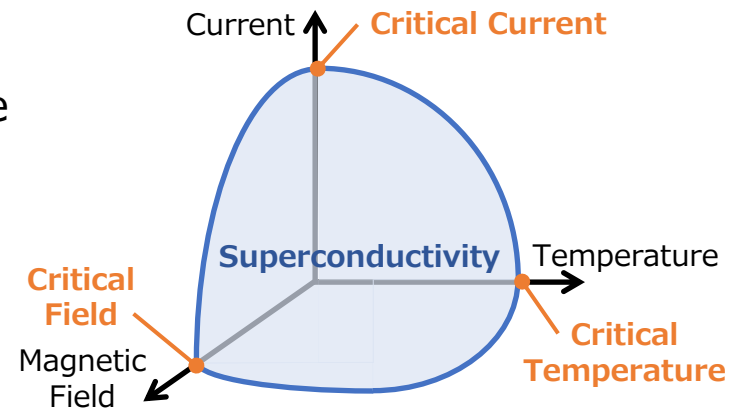
■ Superconductivity

An electrical resistivity becomes exactly zero which occurs in certain materials below a characteristic temperature.

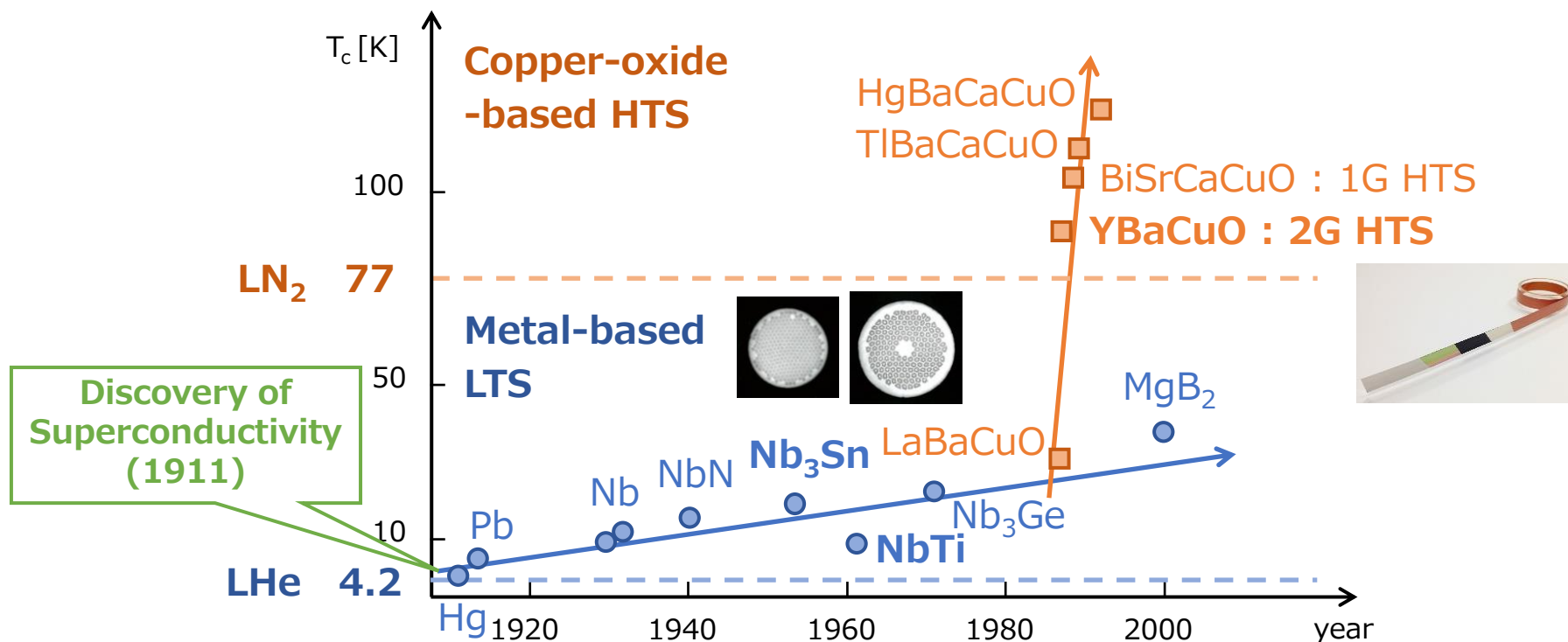


■ Three Critical Points at Superconducting State

- Critical Current --- Maximum current
- Critical Temperature --- Maximum temperature
- Critical Field --- Maximum magnetic field



Historical Discovery of Superconductor



Low Temperature Superconductors (Metal-based)

- Cooling below LHe temperature required
- Practical use in conventional superconducting applications

High Temperature Superconductors (Copper-Oxide-Based)

- Critical temperature higher than LN₂ temperature
 - Verification stage for practical use in industrial applications
- Bismuth (Bi) : 1st generation (1G)
- Yttrium (Y) or Rare-Earth : 2nd generation (2G)**

Historical Development of HTS at Fujikura

The World's First 10m



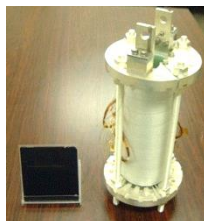
50Ax10m
(2001)

The World's First 200m



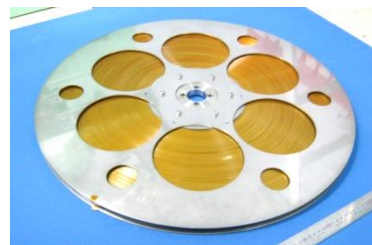
88Ax217m
(2005)

The World's First Cryo-cooled Magnet



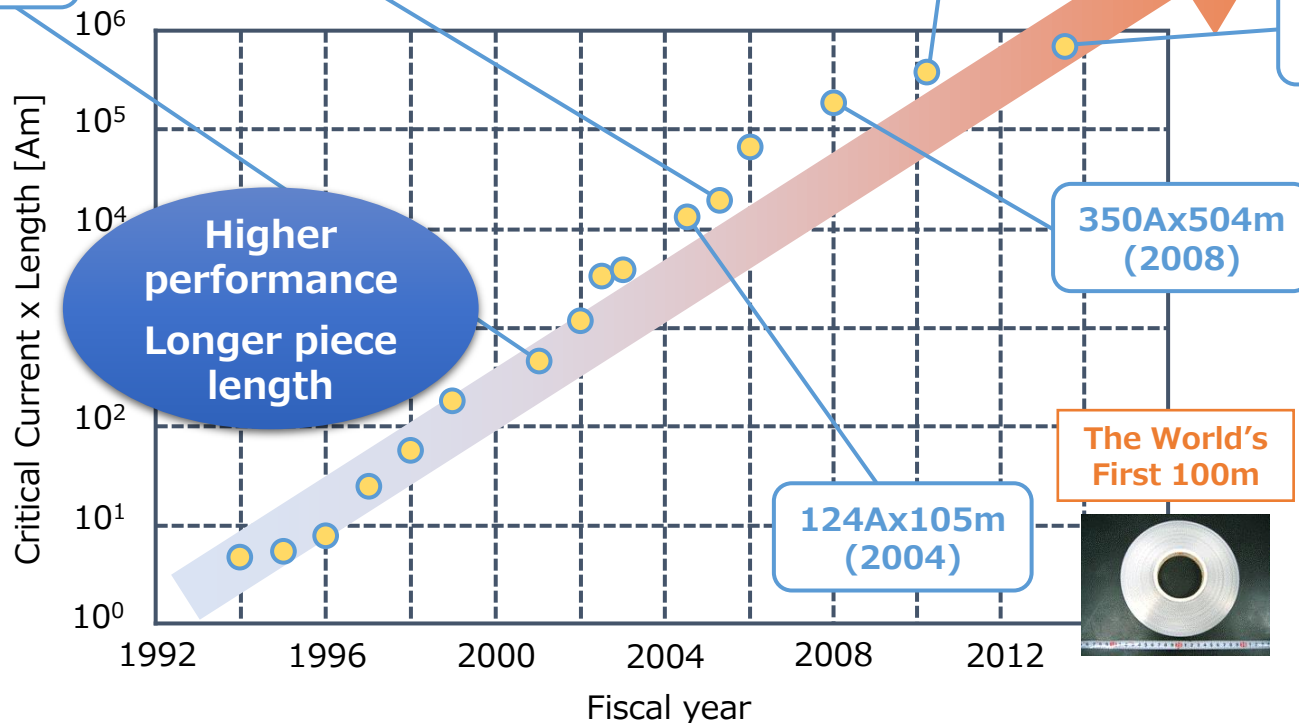
2005
Cryo-cooled magnet

World Record in 2011



572Ax816m
(2011)

More stable
and Higher
productivity



577Ax1040m
(2012)

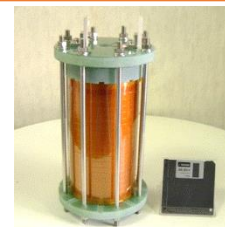
350Ax504m
(2008)



The World's First 100m



The Word's First Solenoid Magnet



2004 LN₂ cooled magnet

■ Typical Specifications

Products	Width [mm]	Thickness [mm]	Substrate [μm]	Stabilizer [μm] ^{*5}	AP Option	Critical Current [A]	
						77K, S.F.	20K, 5T ^{*4}
FYSC-SCH04	4	0.13	75	20	Non-AP ^{*2}	≥ 165	368
FYSC-SCH12	12	0.13	75	20	Non-AP ^{*2}	≥ 550	1,104
FYSC-S12 ^{*1}	12	0.08	75	—	Non-AP ^{*2}	≥ 550	—
FESC-SCH02	2	0.11	50	20	AP ^{*3}	≥ 30	320
FESC-SCH03	3	0.11	50	20	AP ^{*3}	≥ 80	480
FESC-SCH04	4	0.11	50	20	AP ^{*3}	≥ 130	640
FESC-SCH04(05)	4	0.07	50	5	AP ^{*3}	≥ 130	640
FESC-SCH12	12	0.11	50	20	AP ^{*3}	≥ 450	1,920
FESC-S12 ^{*1}	12	0.06	50	—	AP ^{*3}	≥ 450	—

*1 Non-copper stabilizer specification is available in only 12mm-wide for current lead or low thermal conducting applications.

*2 Non-AP specification is mainly for conductors or other general use at relatively higher temperature.

*3 Artificial pinning specification is mainly for use in magnet applications at low temperature and high magnetic field.

*4 I_c at 20K,5T is a reference value and no guarantee of the actual performance.

*5 If requested, an option customizing copper thickness is also available. (e.g., 5μm, 10μm or 40μm)

<Schematic of RE-based HTS tape>

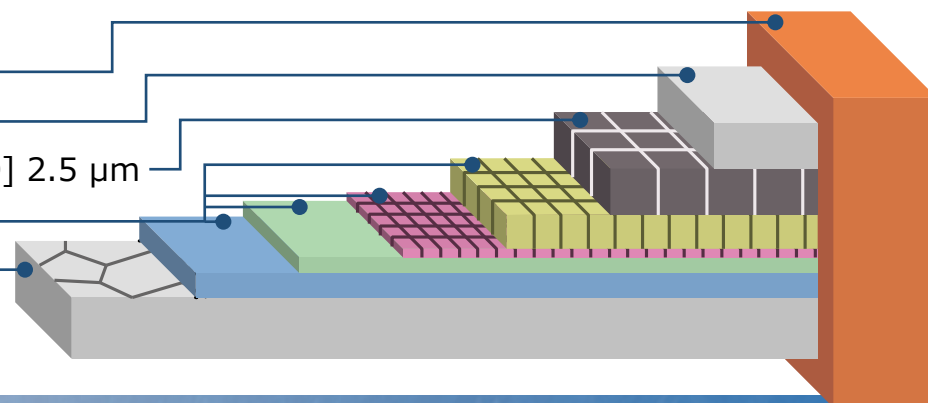
Stabilizer [Cu plating] 20μm

Protection layer [Ag] 2μm

Superconducting Layer [GdBCO] 2 μm / [EuBCO+BHO] 2.5 μm

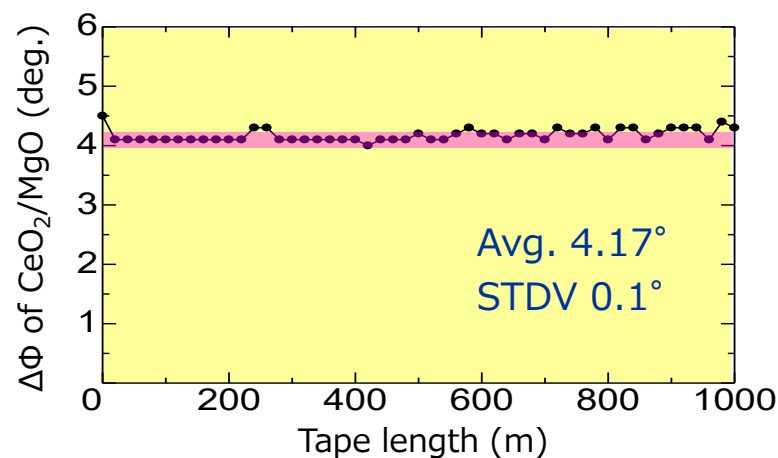
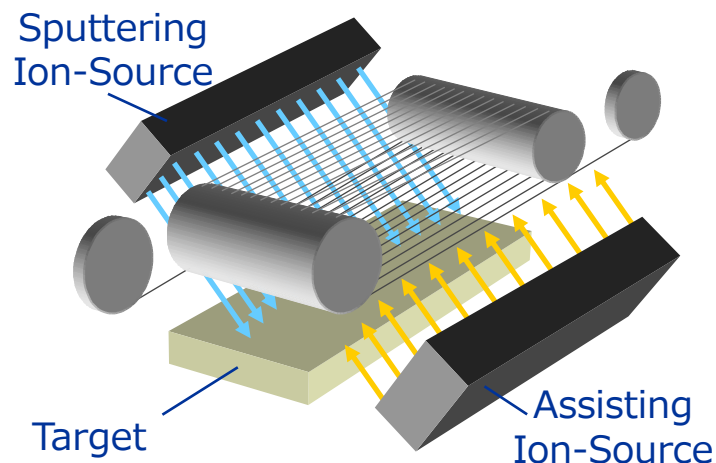
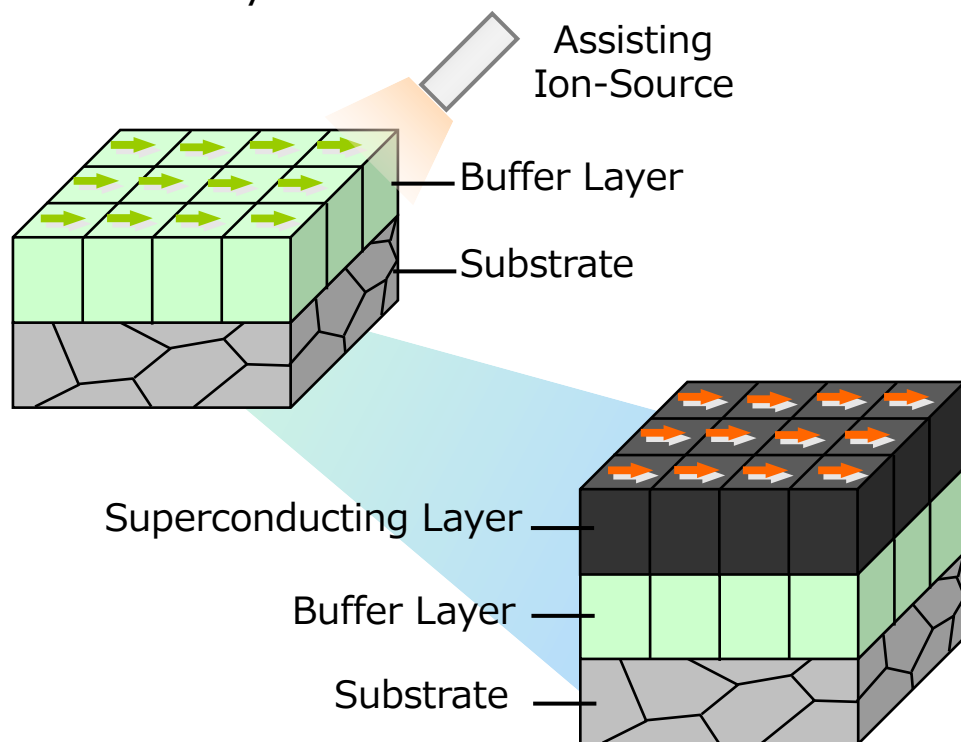
Buffer layer [Al₂O₃ / Y₂O₃ / MgO / CeO₂] 0.7μm

Substrate [Hastelloy®] 75 / 50 μm



■ Ion Beam Assisted Deposition; IBAD

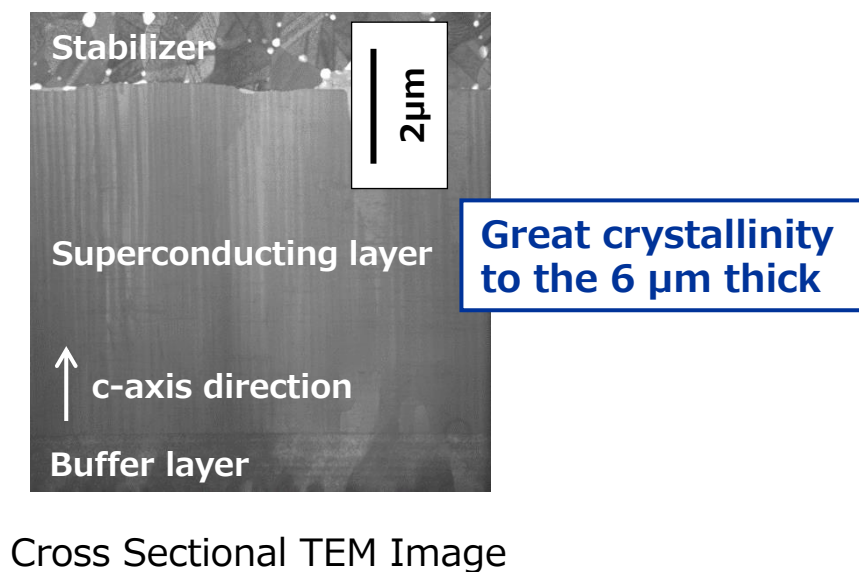
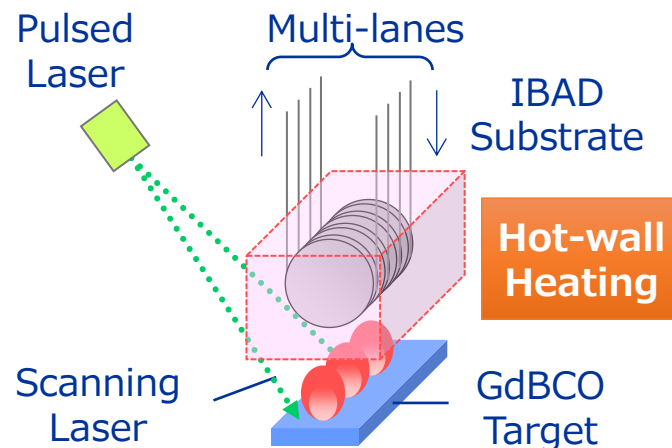
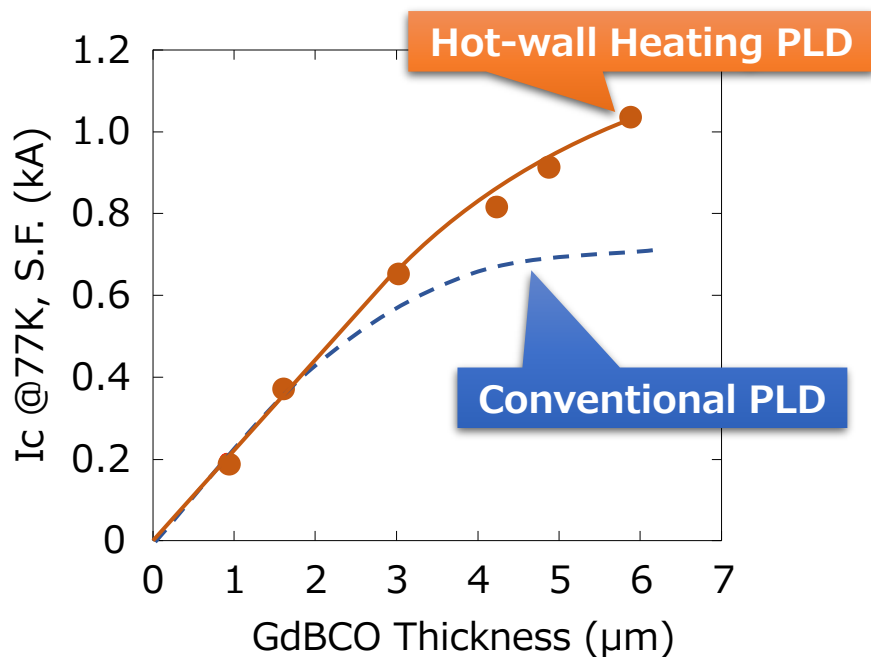
- Fabricating process of buffer layer
- Fujikura original technique (Developed in 1991)
- IBAD enables to fabricate high-textured buffer layer.



1km-length PLD-CeO₂/IBAD-MgO Substrate

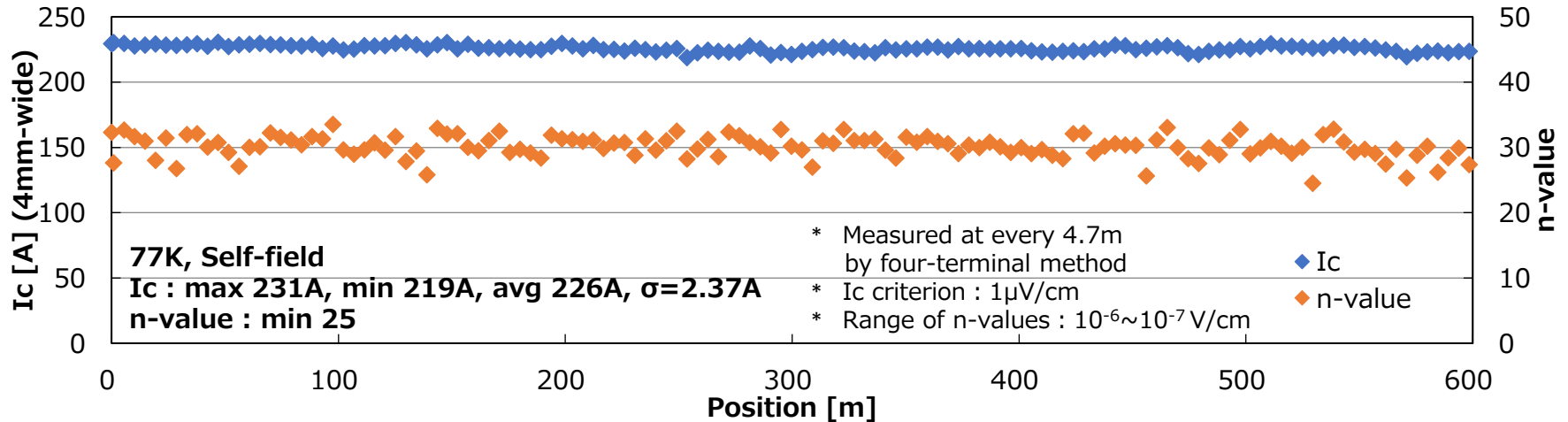
■ Pulsed Laser Deposition; PLD

- Fabricating process of superconducting layer
- Hot-wall heating system is Fujikura original.
- Stable temperature during deposition enables higher superconducting performance than conventional type PLD.

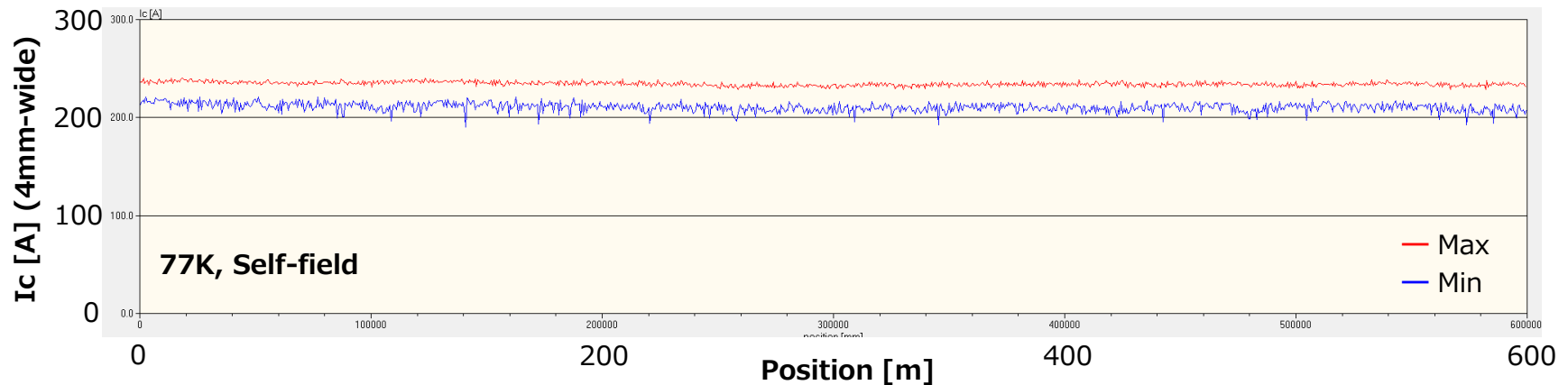


Ic Uniformity of 4mm-wide - Non AP

■ Current conduction measurement - 4mm-wide without AP (FYSC-SCH04)

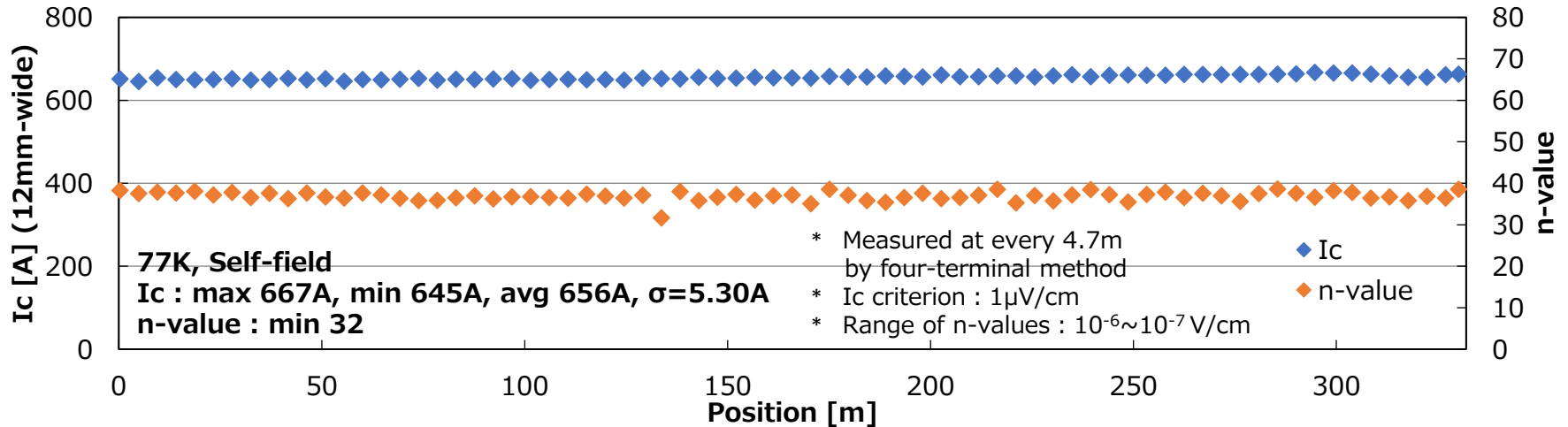


■ Magnetic measurement @Tapestar™ - 4mm-wide without AP (FYSC-SCH04)

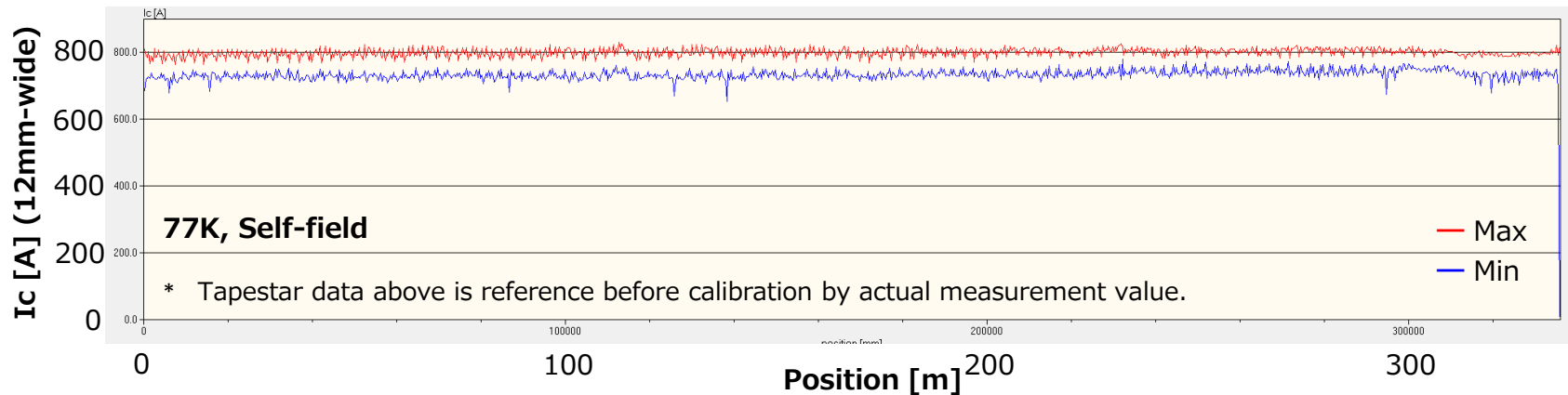


Ic Uniformity of 12mm-wide - Non AP

■ Current conduction measurement - 12mm-wide without AP (FYSC-SCH12)

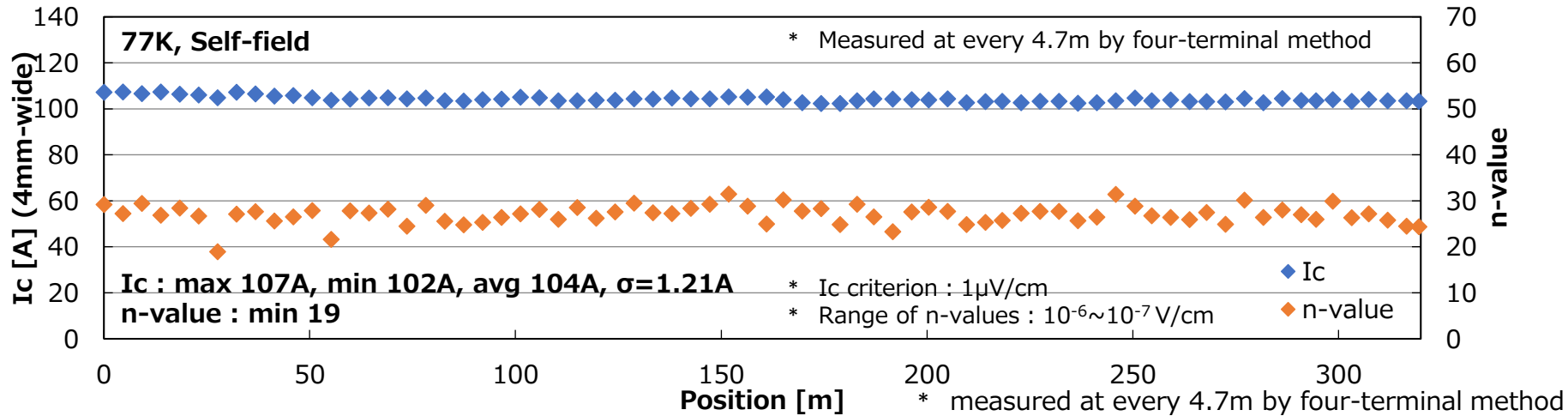


■ Magnetic measurement @Tapestar™ - 12mm-wide without AP (FYSC-SCH12)

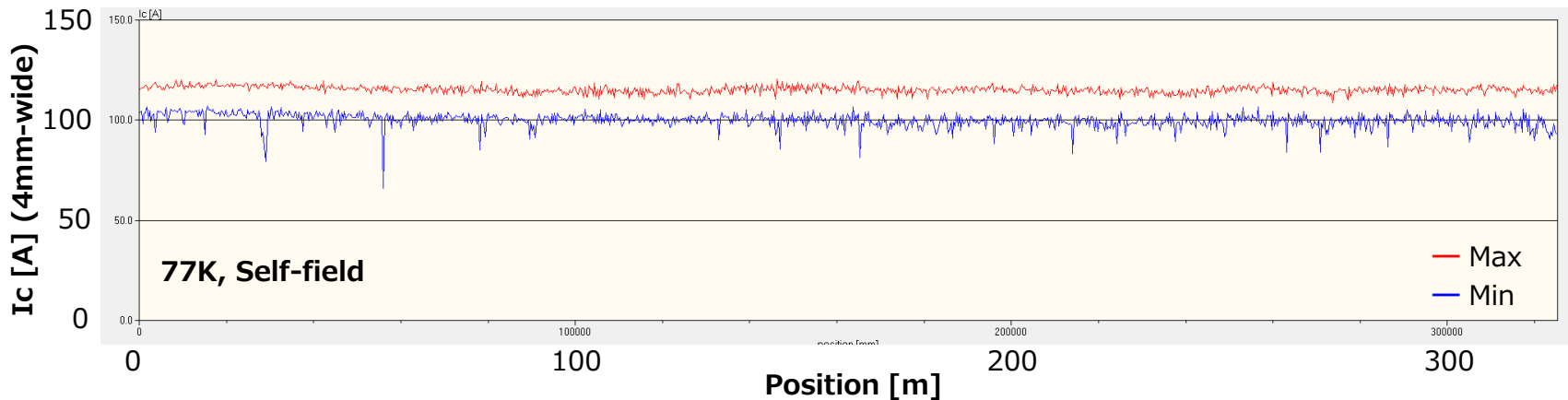


Ic Uniformity of 4mm-wide - AP

■ Current conduction measurement - 4mm-wide with AP (FESC-SCH04)

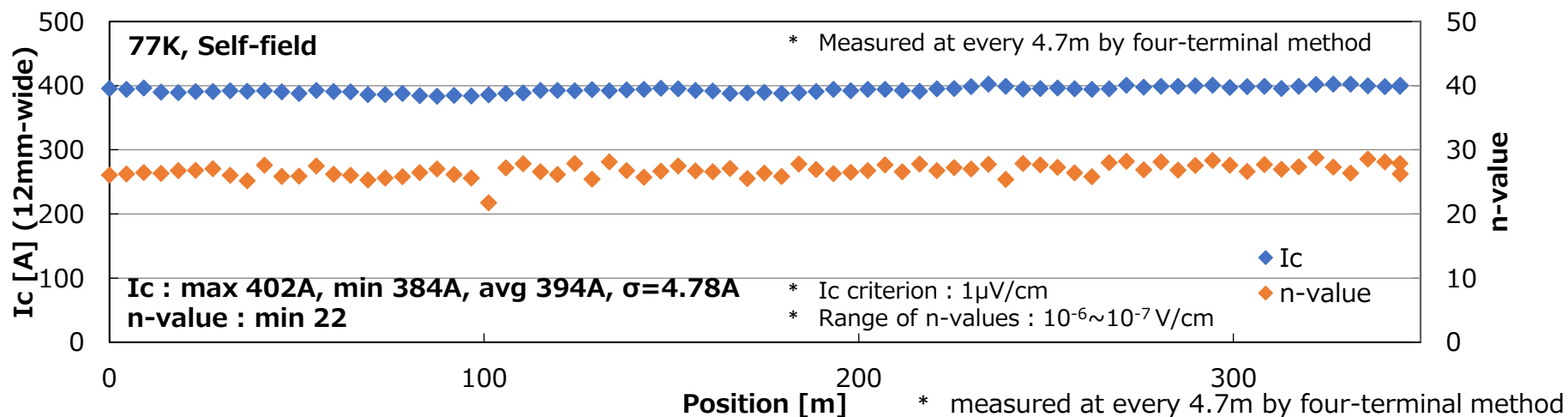


■ Magnetic measurement @Tapestar™ - 4mm-wide with AP (FESC-SCH04)

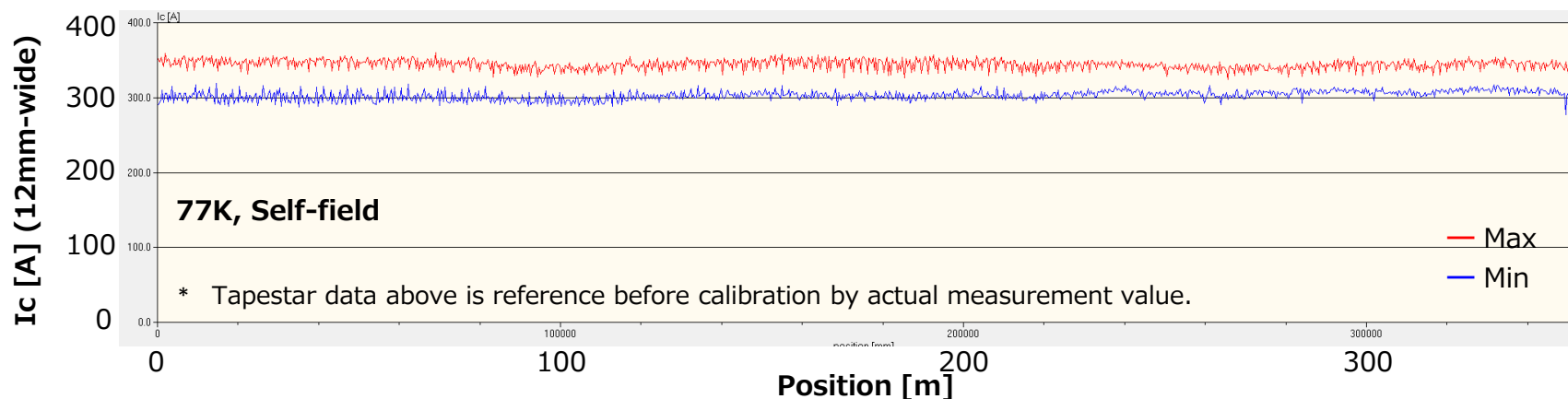


Ic Uniformity of 12mm-wide - AP

■ Current conduction measurement - 12mm-wide with AP (FESC-SCH12)



■ Magnetic measurement @Tapestar™ - 12mm-wide with AP (FESC-SCH12)

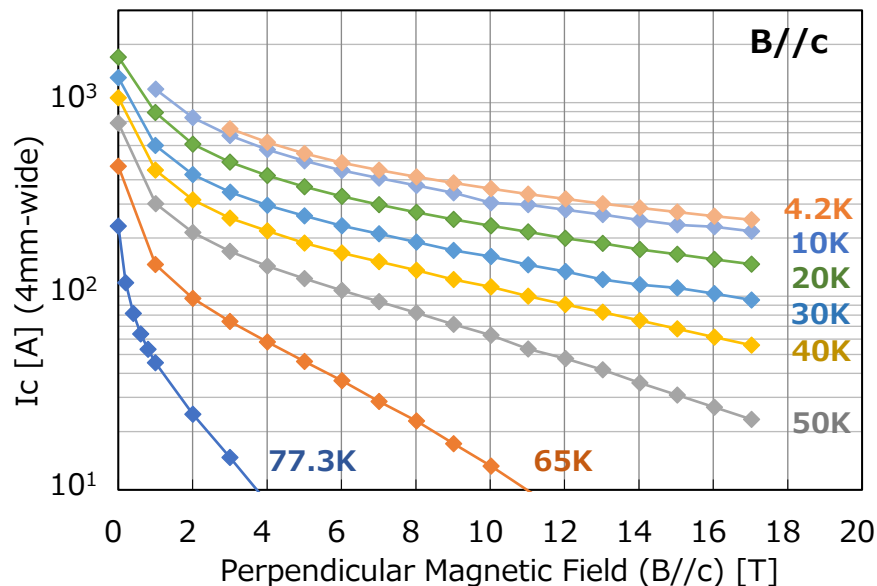


■ Typical in-field I_c performance without artificial pinning (FYSC) (Reference)

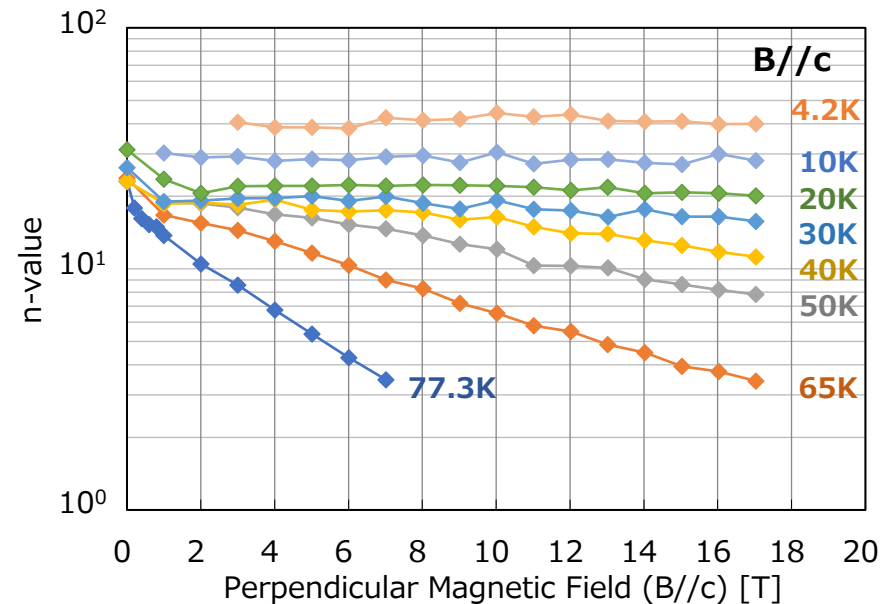
Sample : $I_c = 230\text{A}$ at 77K, Self-field (4mm-wide) (Superconducting layer thickness: $1.9\mu\text{m}$)

➤ Perpendicular Magnetic Field (B//c)

I_c - B - T



n-value - B - T



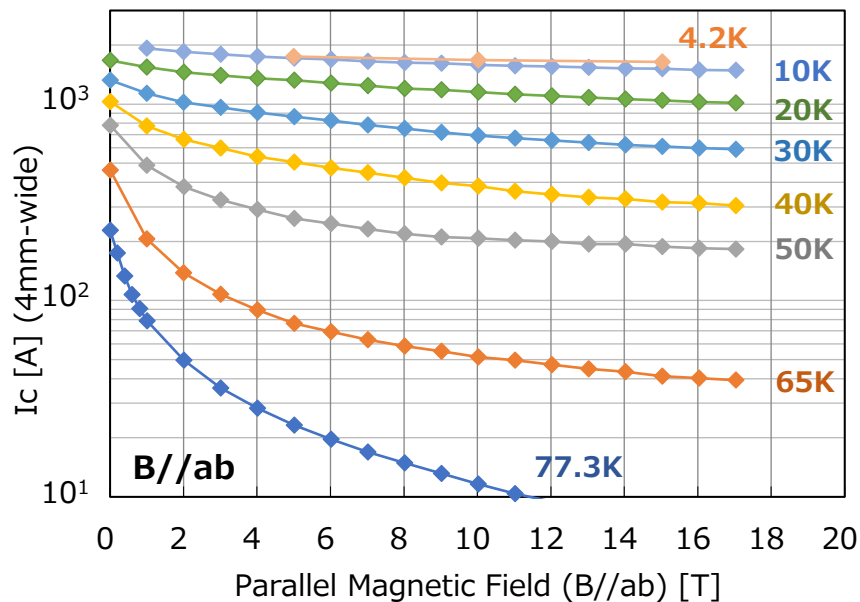
* This work includes some data measured at High Field Laboratory for Superconducting Materials, Institute for Materials Research, Tohoku University.

■ Typical in-field I_c performance without artificial pinning (FYSC) (Reference)

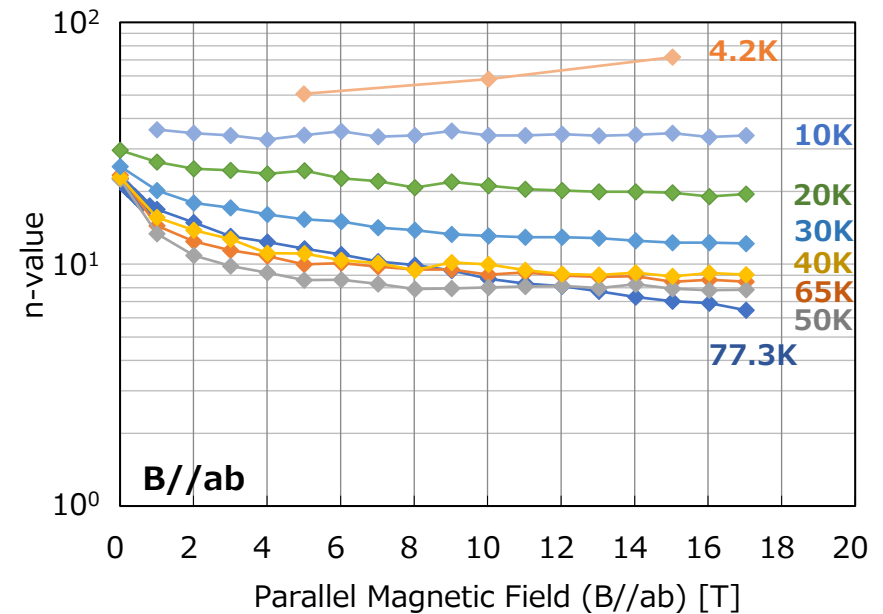
Sample : $I_c = 230\text{A}$ at 77K, Self-field (4mm-wide) (Superconducting layer thickness: $1.9\mu\text{m}$)

➤ Parallel Magnetic Field (B//ab)

I_c - B - T



n-value - B - T



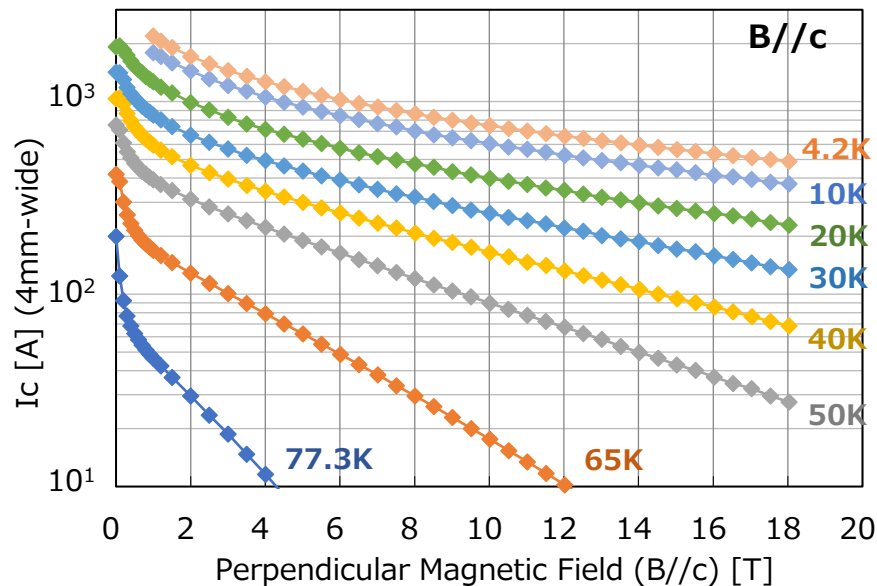
* This work includes some data measured at High Field Laboratory for Superconducting Materials, Institute for Materials Research, Tohoku University.

■ Typical in-field I_c performance with artificial pinning (FESC) (Reference)

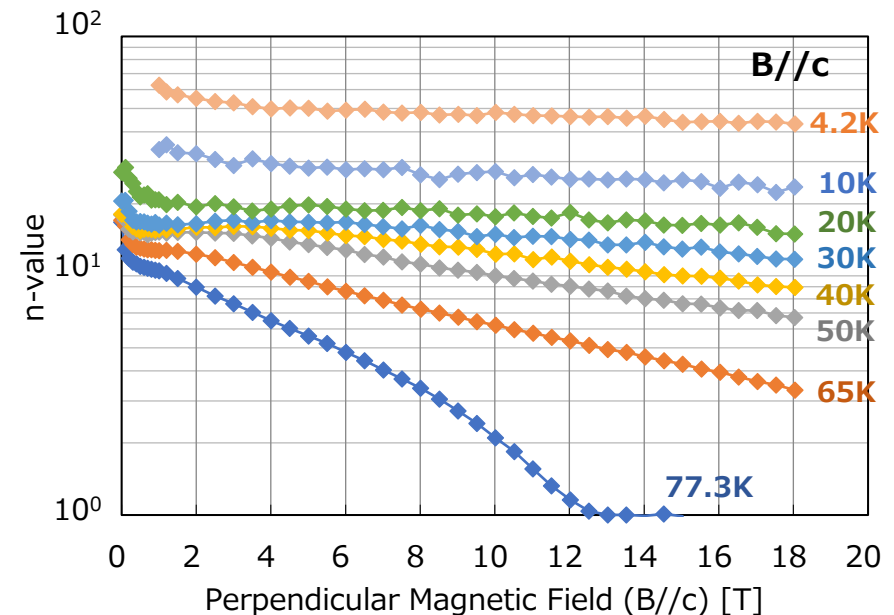
Sample : $I_c = 200\text{A}$ at 77K, Self-field (4mm-wide) (Superconducting layer thickness: $2.4\mu\text{m}$)

➤ Perpendicular Magnetic Field (B//c)

I_c - B - T



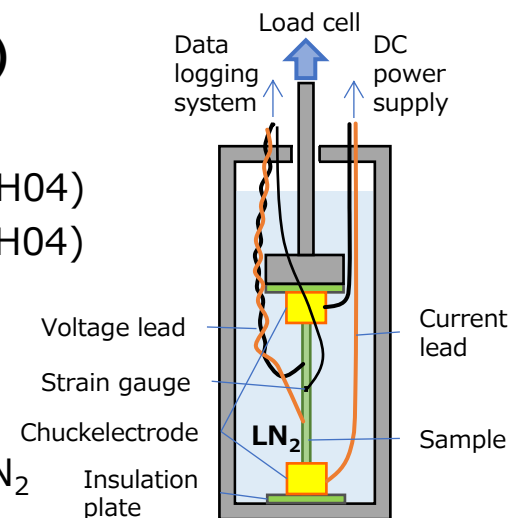
n-value - B - T



* This work includes some data measured at High Field Laboratory for Superconducting Materials, Institute for Materials Research, Tohoku University.

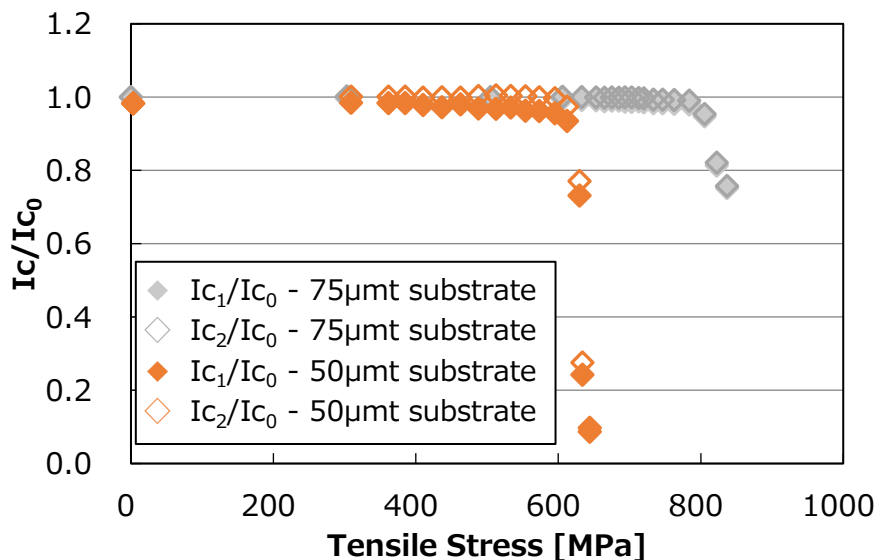
■ Tensile stress evaluation at LN₂ temperature (Reference)

- Sample :
4mm-wide, 75 μm-thick Hastelloy + 20 μm-thick Cu plating (FYSC-SCH04)
4mm-wide, 50 μm-thick Hastelloy + 20 μm-thick Cu plating (FESC-SCH04)
- Measurement method :
1. I_c measurement without load in LN₂ (I_{c0})
2. I_c measurement with applying tensile strain in LN₂ (I_{c1})
3. I_c measurement without load (I_{c2}) after applying tensile strain in LN₂

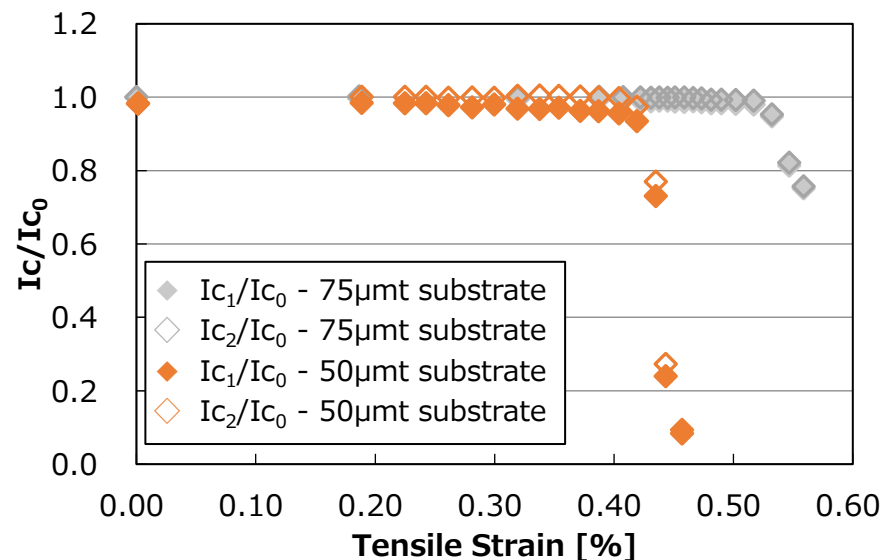


Schematic of tensile test

I_c/I_{c0} versus tensile stress

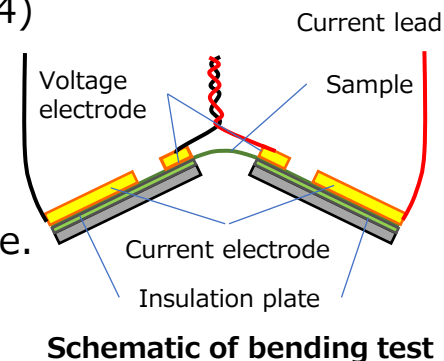


I_c/I_{c0} versus tensile strain



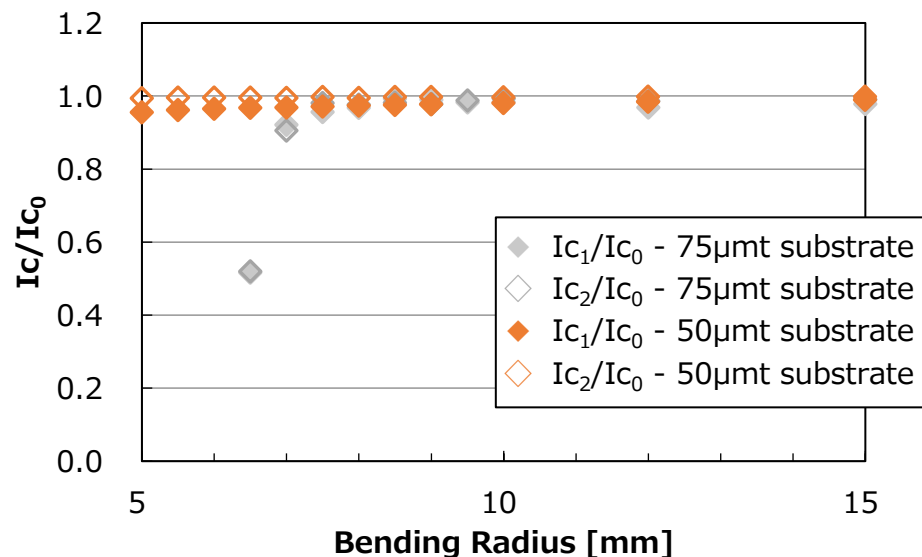
■ Bending property evaluation at LN₂ temperature (Reference)

- Sample :
4mm-wide, 75 μm-thick Hastelloy + 20 μm-thick Cu plating (FYSC-SCH04)
4mm-wide, 50 μm-thick Hastelloy + 20 μm-thick Cu plating (FESC-SCH04)
- Measurement method ("Goldacker" continuous bending method) :
 1. Ic measurement in straight in LN₂ (Ic₀)
 2. Ic measurement with applying bending strain at LN₂ (Ic₁)
 - * Bending direction is tensile direction with superconducting layer outside.
 3. Ic measurement in straight (Ic₂) after applying bending strain in LN₂



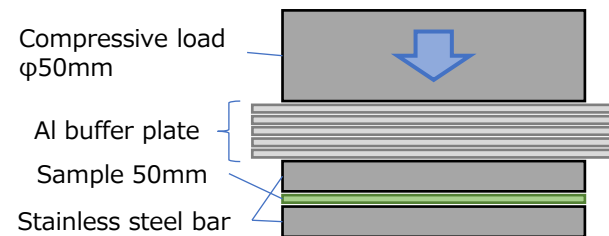
Ic/Ic₀ versus bending radius

**No Ic degradation for
50μmt substrate observed
below the measurement
limit bending radius of 5mm**



■ Compressive stress evaluation in thickness direction at RT (Reference)

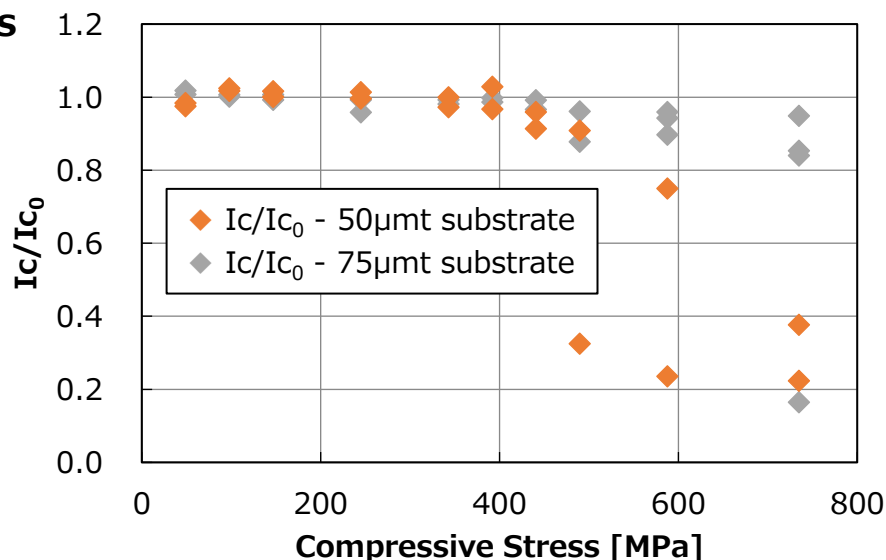
- Sample :
 - 4mm-wide, 75 μm -thick Hastelloy + 20 μm -thick Cu plating (FYSC-SCH04)
 - 4mm-wide, 50 μm -thick Hastelloy + 20 μm -thick Cu plating (FESC-SCH04)
- Measurement method :
 1. I_c measurement in LN_2 (I_{c0})
 2. Apply compressive load in thickness direction of the sample at room temperature
 3. I_c measurement in LN_2 (I_c) after removing compressive load



Schematic of compressive test in thickness direction

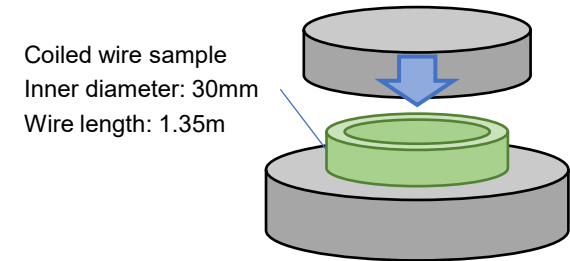
I_c/I_{c0} versus compressive stress

No I_c degradation observed below 400 MPa of compressive stress in thickness direction



■ Compressive stress evaluation in width direction at RT (Reference)

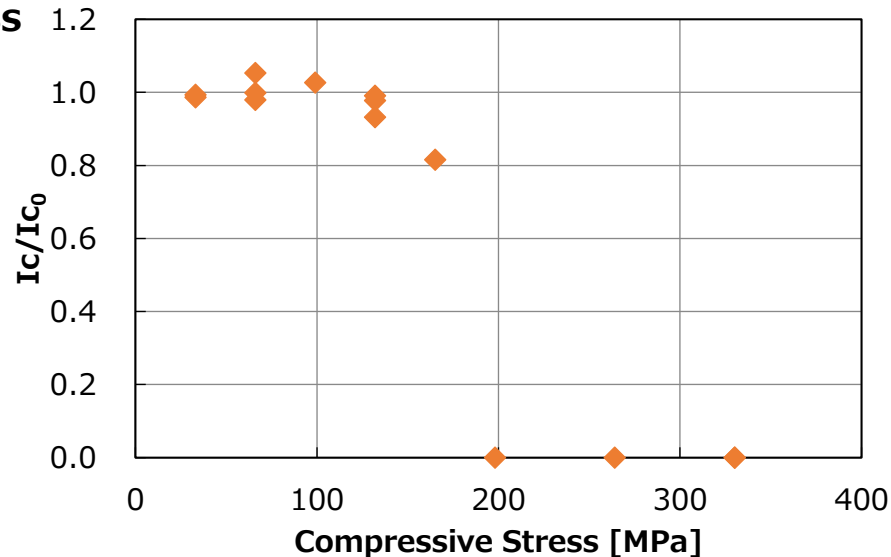
- Sample :
4mm-wide, 50 μm -thick Hastelloy + 20 μm -thick Cu plating (FESC-SCH04)
- Measurement method :
 1. I_c measurement in LN_2 (I_{c0})
 2. Apply compressive load in width direction of the coiled sample at room temperature
 3. I_c measurement in LN_2 (I_c) after removing compressive load



Schematic of compressive test
in width direction

I_c/I_{c0} versus compressive stress

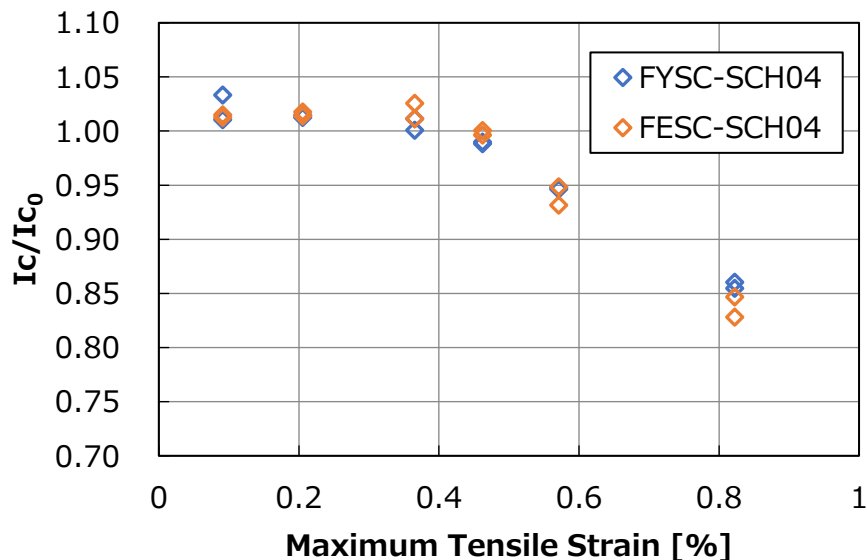
No I_c degradation for
50 μmt substrate observed
below 100 MPa of
compressive stress in width
direction



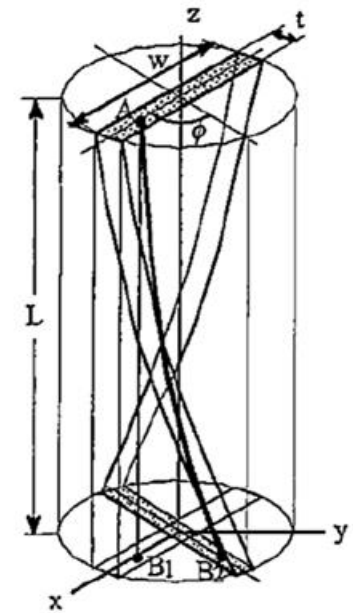
■ Twisting test at room temperature (Reference)

- Sample :
4mm-wide, 75 μm -thick substrate + 20 μm -thick Cu plating (FYSC-SCH04)
4mm-wide, 50 μm -thick substrate + 20 μm -thick Cu plating (FESC-SCH04)
- Measurement method :
1. I_c measurement in straight in LN_2 (I_{c0})
2. Apply twisting load at one side of the sample at room temperature
(Twist pitch=240mm, Tension load=2kgf)
3. I_c measurement in LN_2 (I_c) after removing twisting load

I_c/I_{c0} versus bending radius



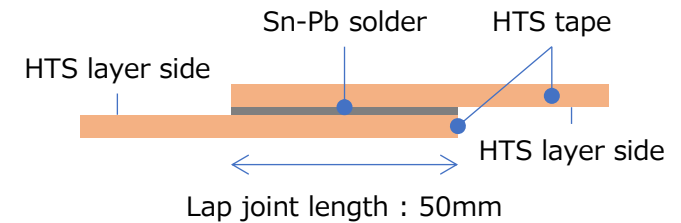
$$\varepsilon = \frac{\phi^2}{2L^2} \left(x^2 - \frac{w^2}{12} \right)$$



M. Takayasu, et al., AIP Conf Proc 1219, 337 (2010)

■ Solder lap joint resistivity (Reference)

- Sample :
4mm-wide, 50 μm -thick substrate + 20 μm -thick Cu plating (FESC-SCH04)
Joint sample with lap joint length of 50mm using Sn-Pb solder, with both superconducting layer sides facing
- Measurement method :
 1. Measure joint resistance in LN_2 (at 77K) with DC of 100A
 2. Calculate a solder lap joint resistivity



Schematic of sample

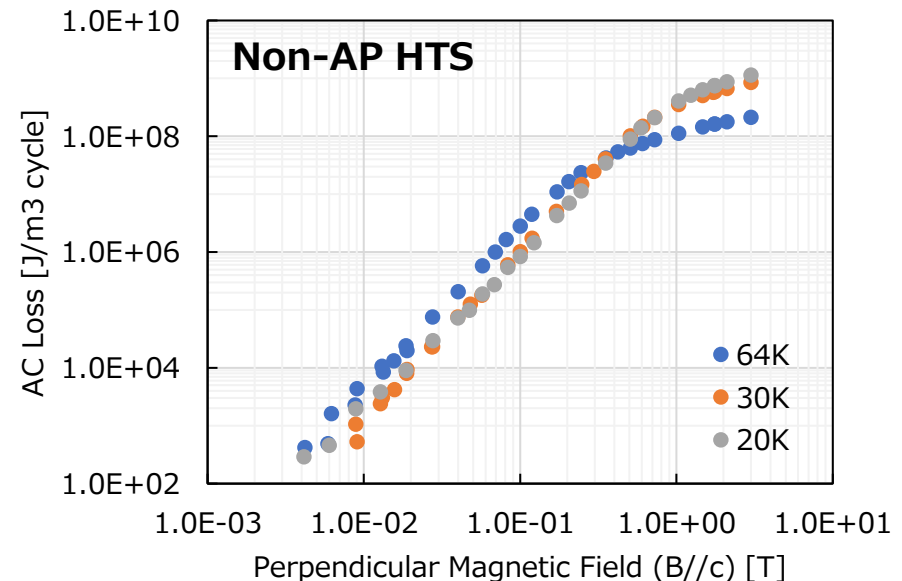
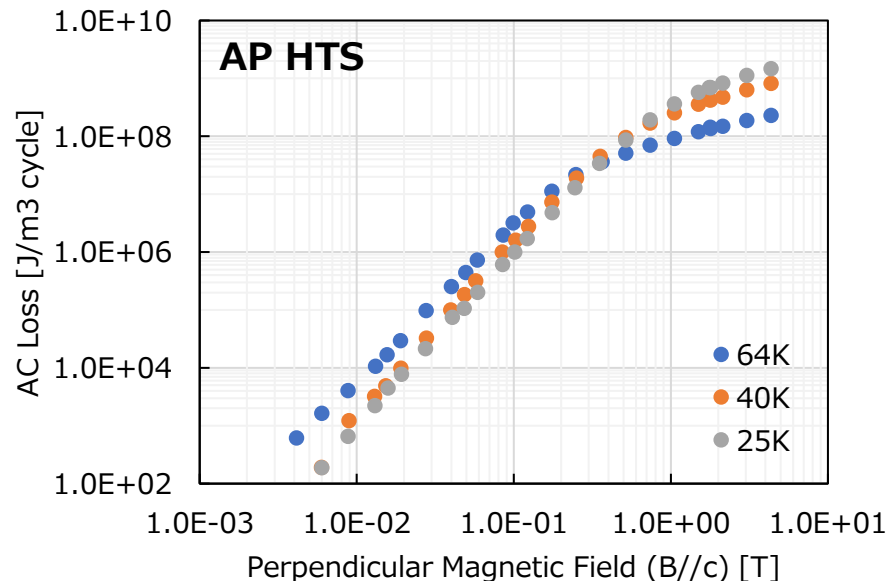
Example measurement results of FESC-SCH04

Sample (n=6)	Joint resistance at 77K (50mm) [n Ω]	Joint resistivity at 77K [n Ωcm^2]
Avg	27.3	54.5
Max	33.6	67.2
Min	21.4	42.8

■ Measurement Condition

Sample	- Artificial pinning (AP; FESC) type, 4mm wide, 50um thick substrate + 20um thick copper plating - Non-artificial pinning (Non-AP; FYSC) type , 4mm wide, 75um thick substrate + 20um thick copper laminate (Old specification)
Temperature	20 - 64 K
Magnetic field (B//c)	0.04 - 4.3 T
Magnetic field frequency	0.01 - 0.2 Hz

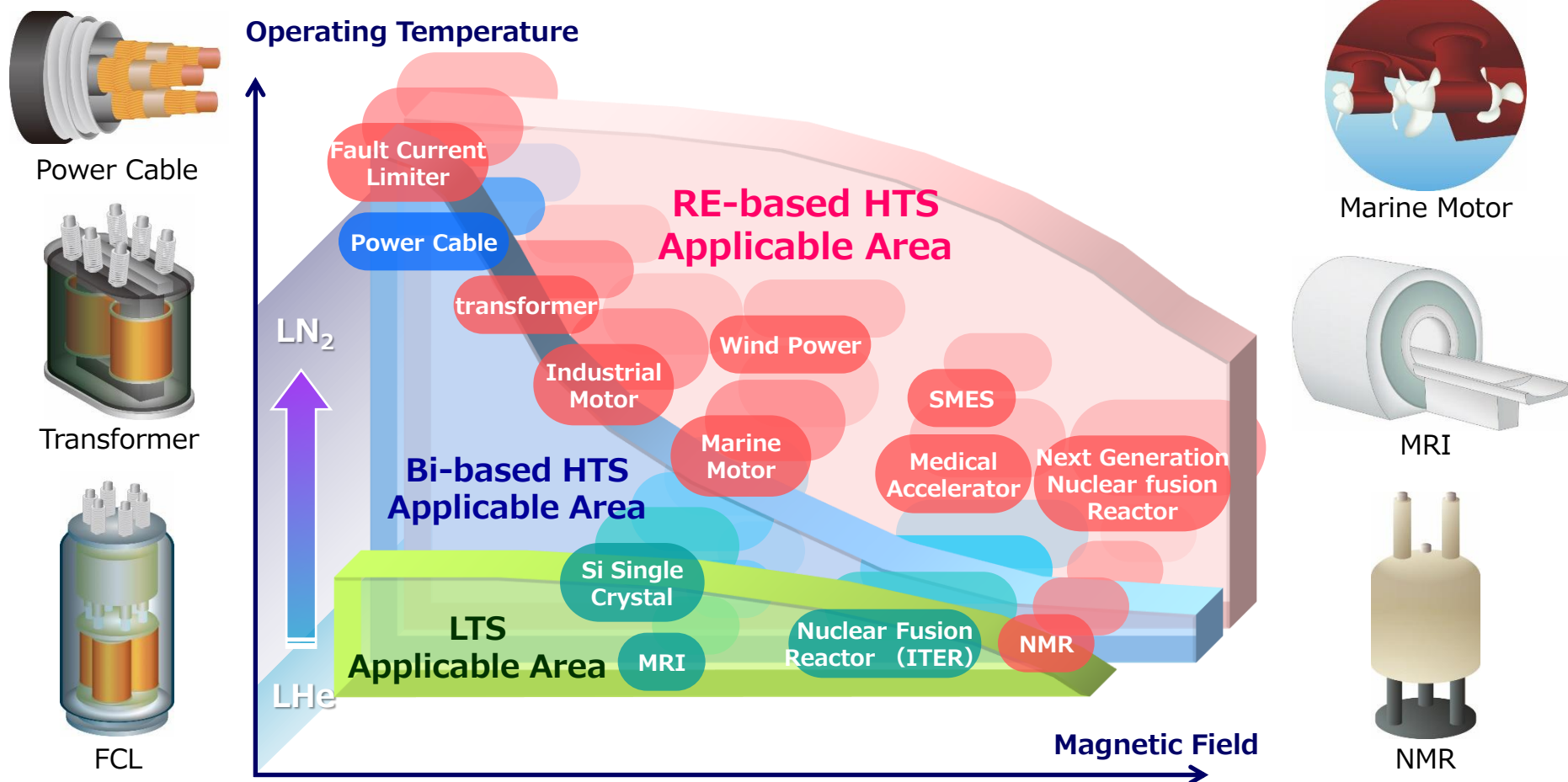
■ Measurement Results



This work includes the data measured at Kyushu University, Japan.

Advantages of Rare-Earth-based HTS

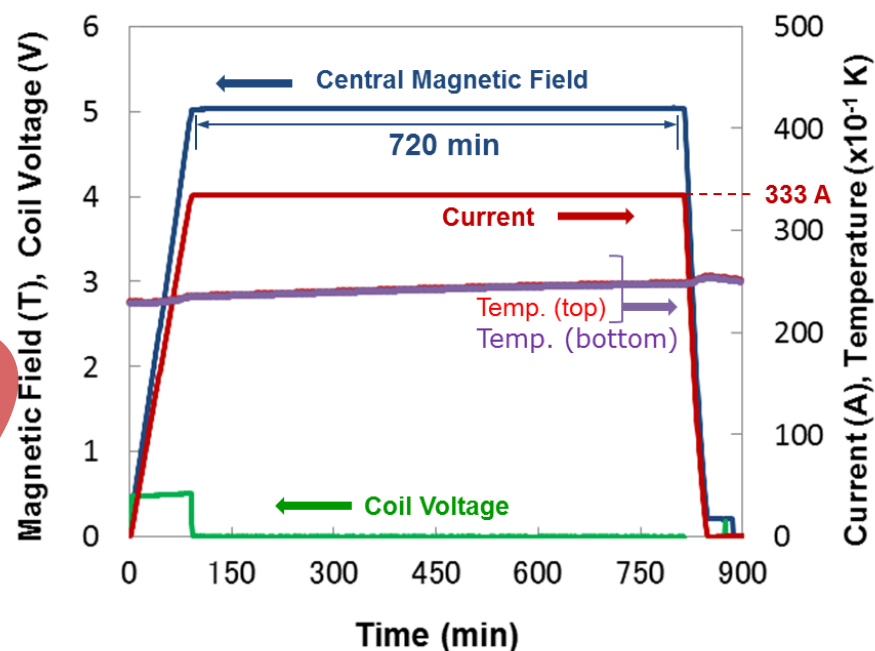
- Higher operating temperature (no use of liquid helium)
- Higher critical current at higher magnetic field
- Smaller size and lighter weight



Development of 5 T 2G HTS Magnet

- Fujikura's 10 mmw Y-based HTS wire
- Total tape length : 7.2 km (300 m x 24)
- Stored energy : 426 kJ

- Composed of 24 pancake coils
- Total number of turns : 5775
- Operating temperature : 25 K



**5 T 2G HTS cryocooled magnet
developed successfully in 2012**

**The magnet excitation performance up to
5 T retained so far after the fabrication**

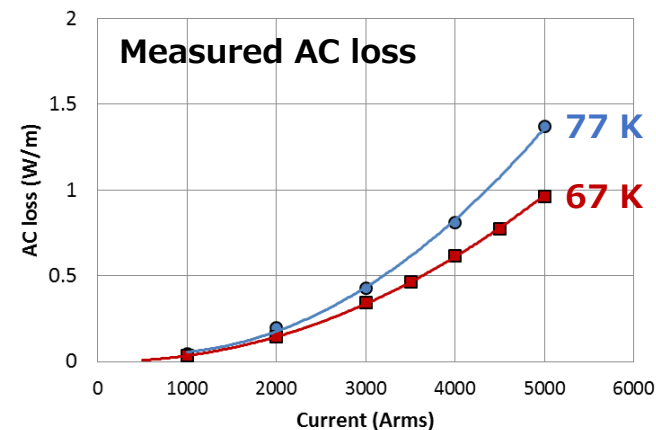
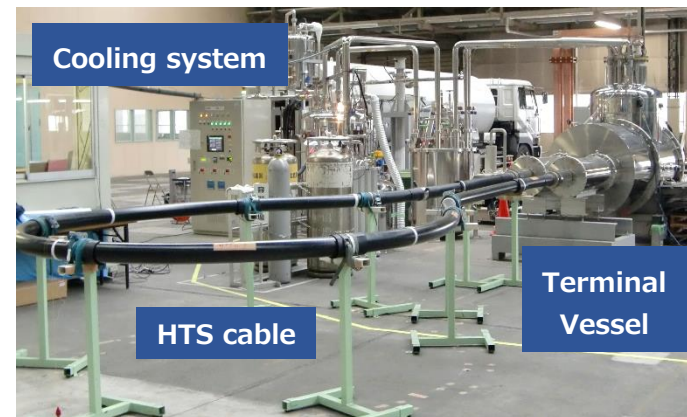
66kV/5kA Class Power Cable (NEDO)

■ Development of HTS power cable with 500A class HTS tape

- Verification of AC Loss reduction with higher I_c HTS
- Single-core in one pipe cable system /66kV-5kA /10m long
- Long term current loading test : 20 cycles
(1 cycle : 8h ON / 16h OFF)
- Target AC loss : < 2 W/m @5kA
Measured AC loss : 1.4W/m@77K, 1.0W/m@67K

< Design and Fabrication >

Items	Specifications
Former	Stranded copper wires (140 mm ²), 20 mmφ
HTS conductor (I _c =14 kA)	4mm-wide wires, 4 layers I _c = 240 A/4 mm-w
Electric insulation	Craft papers (6mm-thick)
HTS shield (I _c =12.7 kA)	All 4mm-w tapes, 2 layers I _c = 240 A/4mm-w
Copper shield	Copper tapes (100mm ²), 44mm
Core protection	non-woven fabric, 45mmφ
Cryostat / Outer sheath	Stainless steel double corrugated pipes with PE jacket, 114mmφ



Fujikura has succeeded in developing RE-based HTS power cable with 5 kA and extremely low AC loss 1.4 W/m in 2013.

This work includes results supported by NEDO

http://www.nedo.go.jp/news/press/AA5_100196.html

■ Recommendable solders

- It shall be generally recommendable to use solders with low melting point and to heat below 200 degrees C within few minutes. In case it would be difficult to melt solder, heating over 200 deg C could be also acceptable with full attentions.
- Pb-free solder could be available with full attention to heating condition. Other solders could be also available depending on application designs or environmental regulation.
- Sn-Bi based or more preferably Sn-Bi-Ag based solder would be recommendable for HTS tapes with silver protection layer such as FYSC-S or FESC-S series. Especially solder including Ag is relatively easy to solder silver protection layer.

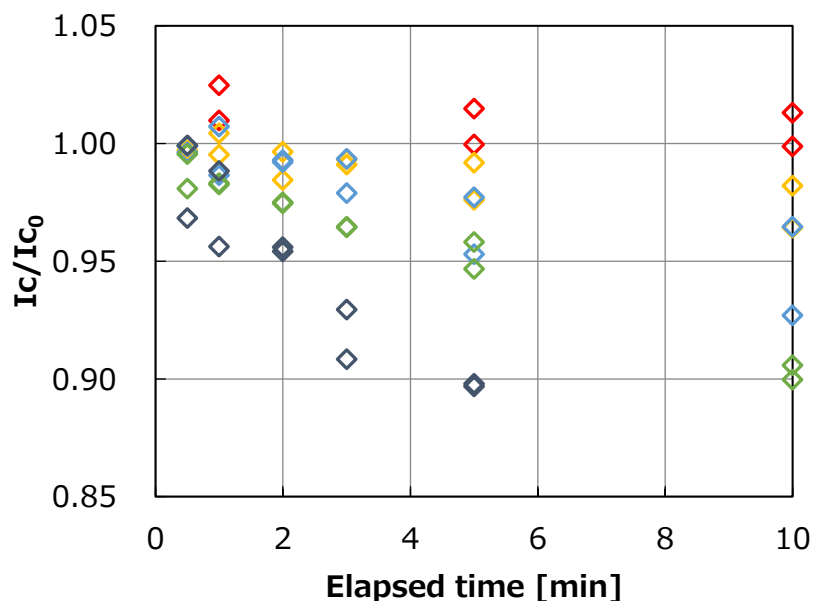
■ Physical properties (Reference)

Solder		Sn-Pb	Sn-Bi	Sn-Ag-Cu	Sn-In	Sn	In
Composition [wt%]		Sn63-Pb37	Sn42-Bi58	Sn96.5-Ag3-Cu0.5	Sn48-In52	Sn (4N)	In (4N)
Melting point [deg C]		183.0	138	217	118	231.9	156.6
Resistivity [nΩm]	297K	167.3	510.6	154.0	168.1	123.1	90.3
	77K	34.6	178.8	19.4	90.3	23.0	17.5

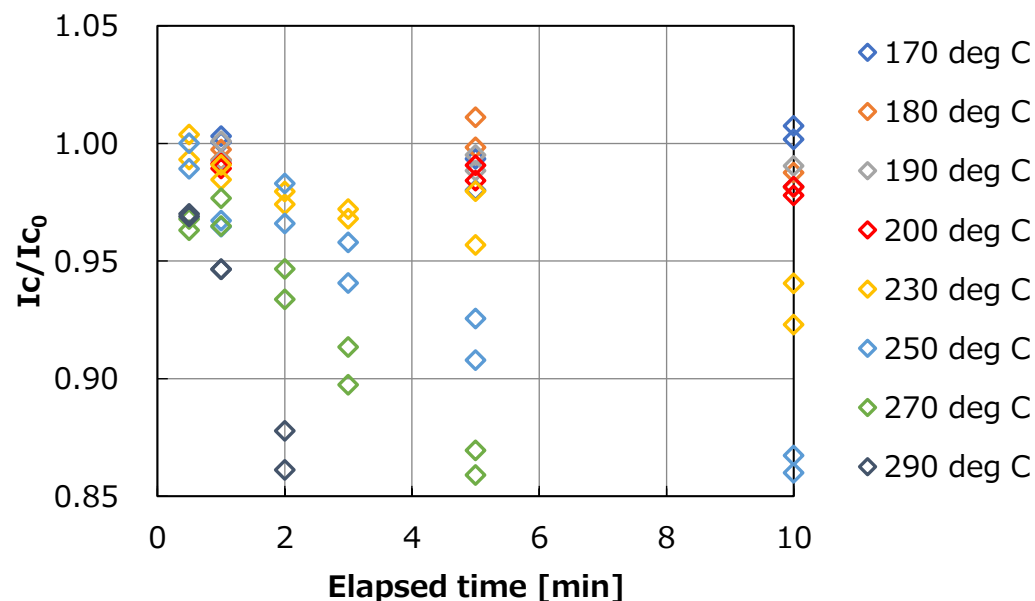
* Resistivity is measured value at Fujikura.

■ Ic degradation during heating

Non artificial pinning (FYSC series)



Artificial pinning type (FESC series)



- * It shall be generally recommendable to heat below 200 degrees C within few minutes. Heating over 200 degrees C could be also acceptable with full attentions to heating condition.
- * These conditions shall not be necessarily applicable to HTS tapes with silver protection layer due to soldering erosion of silver layer.

■ Handling

- Do not exceed the minimum bending radius. The tape will be permanently damaged.
- Do not exceed the maximum tensile stress. The tape will be permanently damaged.
- Do not expose the tapes to moisture.
- Avoid twisting the tapes during spooling.
- The insulating tape may loosen when cut. To avoid this clamp both sides while cutting.
- Do not touch the tapes directly with bare hands. This will cause oxidation and discoloration.
- Do not apply excessive pressure.

■ Storage

- Electroplated copper is susceptible to Oxidation. The storage in sealed conditions are recommendable.
- Store at room temperature, away from heat and moisture.
- Avoid condensation or exposure to corrosive substances.
- Do not place heavy objects on superconducting tapes and reels.

■ Environmental regulation

- All the HTS tapes and packing materials are compliant to RoHS.

■ Export control

- The export of the HTS tapes are controlled under Foreign Exchange and Foreign Trade Law of Japan.

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